

1.5. HARDWARE AND SOFTWARE REQUIREMENTS

To successfully implement and run the online voting system, the following hardware and software components are required:

Hardware Requirements

- **Processor:** Intel Pentium i3 or higher (to ensure smooth performance).
- **Memory (RAM):** At least 4 GB RAM to handle multiple user requests simultaneously.
- **Storage:** Minimum of 500 GB hard drive space to store application files and database records.
- **Monitor:** 15-inch LED display or larger for optimal user experience during development and monitoring.

Software Requirements

- **Operating System:** Microsoft Windows 10 or Windows 11 for compatibility with the development environment and tools.
- **Programming Language:** Java will serve as the core language for backend development, ensuring cross-platform support and strong object-oriented capabilities.
- **Frontend Technologies:**
 - **JSP (JavaServer Pages)** – For dynamic web content generation.
 - **HTML, CSS, JavaScript** – For designing a responsive and user-friendly interface.
- **Database Management System:** MySQL will be used to manage all the data related to voters, candidates, and election results in a secure and organized manner.